

Research Proposal (16 GB VRAM · 10-Day Feasible Edition)

What Sounds Break a Speech-LLM's Ear — and For Whom?

A descriptor-level, fairness-aware audit of Large Audio-Language Models under controlled background interference

AIP-Speech — Acoustic Interference Profiling for Speech-LLMs. Inference-only · **single 16 GB GPU** · ~10 days · **< 3 GB storage** · stimuli mixed on-the-fly but **fully inspectable and reproducible** (§3).

Metadata

Field	Value
Project	AIP-Speech — Acoustic Interference Profiling for Speech-LLMs
Researcher	Masaru — IvLabs
Venue	IMPACT-SPEECH @ EMNLP 2026 (archival short paper, 4 pp ACL; long-paper extensible)
Deadline	15 July 2026 (AoE) — <i>re-confirm on the official CFP</i>
Compute	Inference-only, single 16 GB GPU , no training; target $\approx$ 85-115 k forward passes
Storage	< 3 GB (source corpora + manifest + JSONL outputs; no persisted stimulus set)
Timeline	10 days , Day-6 go/no-go gate, robustness-null fallback

No scene-mapping anywhere. There is no "hospital/office/classroom." Every background is a **named, public, downloadable recording** characterized by **measurable acoustic descriptors**, and every foreground dataset is stated inline with the task that uses it (§4). The only construction step is **SNR mixing per a released manifest** — standard noise-robustness practice — with **no ad-hoc synthesis** of babble (§3.2).

1. The question, in one paragraph

Holding the spoken content fixed and mixing a real background recording behind it at a controlled signal-to-noise ratio, we ask: **which measurable properties of a background sound determine how badly a Speech-LLM fails, is that failure equitable across speakers, and is it driven by the sound's *meaning* or merely its *loudness*?** We answer with two objective tasks (ASR, keyword spotting), ~20 descriptor-annotated real backgrounds, five 16 GB-feasible models, and a scoped set of controls.

2. Hypotheses

ID	Hypothesis	Settled by Experiment
H1 — Profile law	Degradation is predicted by background <i>descriptors</i> beyond SNR; speech-likeness, linguistic content, 2-8 Hz (syllabic) modulation dominate.	E1
H2 — Semantic, not energetic	The harm of speech-like backgrounds shrinks under a spectrally-matched, phase-scrambled twin at the same SNR.	E2
H3 — Content injection	LALMs leak background content into outputs (ASR insertions; KWS false triggers), above the determinism floor, concentrated on linguistically-loaded backgrounds.	E2
H4 — Disparate robustness	The noise penalty differs across accent/gender; speech-like interference widens the gap more than stationary noise.	E3
H5 — Implicit (stretch)	"Ignore background sound" reduces but does not eliminate the effect.	E4
H0 — Robustness null	Degradation is a pure function of SNR, equitable across subgroups, no injection above the floor.	all — a clean, publishable result

3. Two design decisions you asked about

3.1 On-the-fly mixing is not opaque — three ways to see every intermediate step

We do not persist millions of mixed WAVs (that is the storage blow-up). But the pipeline stays fully inspectable:

- Deterministic regeneration on demand.** Every stimulus is a pure function of `(speech_id, background_id, snr, condition, seed)`. A one-line `materialize(row)` call re-creates the *identical* waveform any time you want to listen to, plot, or hand-check it. Nothing is lost — it is recomputed bit-for-bit.
- A persisted inspection sample.** A fixed ~60-clip sample spanning tasks × backgrounds × SNRs *is* written to `checks/inspection/` for manual listening and for all manipulation checks (background-presence, WER-constancy, scramble-validity).
- A per-stimulus diagnostics log.** At mix time we log, *for every stimulus*, the numbers that say "what happened" — achieved SNR, pre/post loudness (LUFS), clipping flag, measured background-presence score, Whisper WER on the mix — to `checks/mix_diagnostics.csv`, **without** saving the audio. You get a complete numeric trace of the whole run plus the ability to regenerate any clip behind any row.

So the storage win costs you nothing in observability: mix-on-the-fly for the bulk, **regenerate-any-clip-exactly** for inspection, and a full diagnostics table for monitoring.

3.2 No synthesized babble — real, citable noise corpora instead

You were right that summing LibriSpeech speakers into "babble" is hard to reproduce and cite. The **interference battery (the manipulated variable, shared by all experiments)** is therefore built **only from named public recordings**:

Battery slice	Source corpus (citable, downloadable)	Why
Diverse non-speech events (~12)	ESC-50 (50 classes, 5 s, CC) — e.g., rain, sea waves, engine, vacuum, footsteps, fire, helicopter, clock-tick, keyboard, dog, rooster, wind	spans the non-speech descriptor space
Real multi-talker babble / speech-like (~5)	DEMAND environments PCAFETER (cafeteria), PRESTO (restaurant), SPSQUARE (public square), OMEETING (meeting); NOISEX-92 <i>babble</i> (canteen, the canonical babble)	the crucial speech-like extreme, recorded, not synthesized
Speech / hubbub / music anchors (~3)	MUSAN speech (read-speech + hubbub) and music subsets	speech-like + music-with-vocals extremes
Stationary anchor (~1)	MUSAN noise / DEMAND TCAR	broadband stationary control

Injection probes (a background that *says a specific word*, for the KWS trigger test) use a **fixed, published list of held-out Google Speech Commands clip IDs** mixed at a fixed SNR — fully reproducible from the released manifest, no synthesis. The released artifact includes exact source IDs + seeds, so every stimulus is regenerable by anyone.

What we measure about each background (descriptors). Every battery clip carries a pre-computed vector — all measurable, none hand-labelled for "scene":

Descriptor	Extractor	Hypothesized role
speech_likeness	P(speech) from a VAD/speech classifier on the background alone	↑ harm
linguistic_content	confidence-weighted word count from Whisper on the background alone	↑ injection
mod_2to8Hz	temporal-envelope modulation energy, 2-8 Hz (syllabic rate)	↑ masking
spectral_overlap	energy fraction in 300-3400 Hz	↑ masking
harmonicity, stationarity, onset_density, loudness	HNR, inverse spectral flux, onsets/s, integrated LUFS	shape vs energy controls

For the ~8 speech-like / semantically-loaded backgrounds we also build a **phase-scrambled, spectrally-matched twin** (same magnitude spectrum + energy, no structure or meaning) — the control at the heart of E2.

4. Experiments

Each experiment lists its tasks; each task states its **foreground dataset (exact portion) × the interference battery (§3.2) at the stated SNRs**, and its **metrics (standard + any custom one we introduce)**. SNR grid is {+10, +5, 0} dB throughout (reference +10; 0 dB is the stress point). The clean (background-free) rendering of each foreground item is the within-item paired baseline. Decoding is greedy/deterministic, so significance comes from **paired permutation tests across items** (a one-time determinism check, not a jitter-floor protocol, sets the sanity bar).

E1 — Interference Profiling: *which sound properties break which task?* (C-PROFILE → H1)

Task 1.1 — ASR under background. - **Data:** LibriSpeech test-clean + test-other (~100 utterances, balanced) × full interference battery × SNR {+10,+5,0}; plus each utterance's clean rendering. - **Metrics (standard):** WER, CER (jwer); substitution/deletion/insertion split. - **Analysis:** mixed-effects regression of ΔWER on descriptors (§6.1). No new metric needed.

Task 1.2 — Keyword spotting / wake-word under background. - **Data:** Google Speech Commands v2 (~120 clips spanning target + non-target words) × full battery × SNR. - **Metrics (standard):** Accuracy; **False-Alarm Rate** (FAR) on non-target backgrounds; Miss rate.

E1 output: the descriptor law — a ranked, signed statement of which background properties drive failure, pooled across both tasks and all models.

E2 — Semantic vs Energetic: *the words or the loudness?* (C-INJECT → H2, H3)

Task 2.1 — ASR: real vs scrambled. - **Data:** LibriSpeech subset × {the ~8 speech-like backgrounds **and** their phase-scrambled twins} at **0 dB only** (the worst, most informative SNR). - **Metrics:** ΔWER(real) − ΔWER(scrambled) (standard WER, contrast design); **custom — Background-Injection Rate (BIR):** fraction of *inserted* tokens in the hypothesis that match the background's own transcript/label vocabulary (from the §3.2 linguistic extractor). A positive real-minus-scrambled gap is the evidence that the effect is *semantic*, not energetic.

Task 2.2 — KWS: trigger injection. - **Data:** Google Speech Commands v2 × {injection-probe backgrounds = held-out Speech Commands clips uttering the target word, fixed published IDs} **and** their scrambled twins, at 0 dB. - **Metrics: custom — Trigger-Injection Rate (TIR):** FAR specifically on the injection probes (the model "hears" the background word and fires), compared to the scrambled twin and to a generic-noise baseline.

E2 output: proof that speech-like backgrounds hurt *because they are words* — the differentiator from "noise hurts ASR."

E3 — Disparate Robustness: *for whom is it worse?* (C-FAIR → H4) — the IMPACT-SPEECH spine

Task 3.1 — ASR fairness, ecological (near-free). - **Data:** Common Voice v17 English (streamed; accent/gender/age-labelled), ~80 utterances chosen to span ≥4 accent groups × both genders × full battery × SNR. Because demographics are *baked into the E1-style ASR run*, this is largely a **post-hoc subgroup cut**, not a separate grid. - **Metrics:** per-subgroup ΔWER; **custom — Robustness Gap** = max_g ΔWER_g − min_g ΔWER_g; **Disparate-Robustness Index (DRI)** = gap / mean Δ.

Task 3.2 — ASR fairness, content-controlled. - **Data:** Speech Accent Archive (everyone reads the *same* "Please call Stella" paragraph), ~6 accent groups × balanced gender (~90 speakers) × {the 8 speech-like + 4 non-speech backgrounds} at **0 dB**. Same words across accents = a clean Δ comparison free of content confounds. - **Metrics:** content-matched per-accent ΔWER; Robustness Gap; **descriptor × subgroup interaction** test (does speech-like widen the gap more than stationary?). Clean-baseline gap overlaid so the *added* inequity from noise is explicit.

E3 output: the bias finding with an explanatory account of *which interference profiles* drive inequity.

E4 — Steerability (stretch): *can you tell it to ignore the background?* (C-STEER → H5)

Task 4.1 — ASR + KWS with a blinding instruction. - **Data:** a 1/3 subset of the E1 stimuli at 0 dB, re-run with the prepended instruction "*Ignore any background sounds; respond as if the audio were recorded in a silent room.*" - **Metrics: custom — Residual-Effect Ratio (RER)** = effect_with_instruction / effect_without (per task/model); instruction-compliance rate.

5. Models (validated for 16 GB)

Five core models spanning the encoder-coupling spectrum, all fitting 16 GB in **understanding-only mode** (speech-generation/talker heads disabled) with **4-bit weights** where needed. Load one at a time; free the GPU (`del model; torch.cuda.empty_cache()`) before the next.

Model	Size / config for 16 GB	Coupling	Role
Qwen2.5-Omni-3B	~3B, fp16 or 4-bit (Thinker only)	end-to-end	SOTA-family, comfortably fits
Qwen2-Audio-7B	4-bit (~5–6 GB)	end-to-end (Whisper-init)	acoustically-sensitive exemplar
Phi-4-multimodal	~5.6B, 4-bit/8-bit	LoRA adapter	strong LLM-decoder ASR, intermediate coupling
Gemma 3n-E4B	effective ~4B, edge-optimized	USM + Gemma	user-requested; fits easily; ≤30 s clips
Kimi-Audio-7B	4-bit	end-to-end	independent pipeline (force task; it defaults to ASR)
(swap/stretch) MiniCPM-o 2.6 / Step-Audio 2 / Audio Flamingo 3	4-bit	—	extended set, reduced grid if ahead
(optional API ~\$10) GPT-4o-Audio / Gemini-Flash	—	closed	reference points

16 GB practicalities: batch size 1–2 for 7B (ASR generation is the bottleneck); KWS clips (~1 s, 1-token outputs) batch larger and dominate throughput; **dry-run a VRAM check** per model before committing it; avoid the full-omni Qwen2.5-Omni-7B (use the 3B or Thinker-only). If a 7B model OOMs even in 4-bit, drop it for a 3B/edge alternative — the roster has slack.

6. Analysis

6.1 C-PROFILE (E1)

Per task: $\Delta \sim \text{SNR} + \text{speech_likeness} + \text{linguistic_content} + \text{mod_2to8Hz} + \text{spectral_overlap} + \text{stationarity} + \text{onset_density} + (1|\text{item}) + (1|\text{model}) + (1|\text{background})$ (`statsmodels / pymer4`). Standardized coefficients + bootstrap CIs + variance-inflation; dissociate collinear descriptors using the real anchor extremes (NOISEX babble vs MUSAN stationary). **Money plot:** Δ vs speech-likeness and vs 2–8 Hz modulation.

6.2 C-FAIR (E3)

Robustness Gap + DRI per task; descriptor×subgroup interaction; clean-baseline gap overlaid.

6.3 C-INJECT (E2)

ASR error taxonomy (sub/del/ins/**injection**) + BIR/TIR; tie injection to linguistic_content; real – scrambled at 0 dB.

6.4 Decision rule

An effect is **real** iff (a) it survives a paired permutation test ($p < 0.05$, corrected across backgrounds) and (b) for semantic claims is **not reproduced by the scrambled twin**, or for energetic claims is **monotone in SNR**. Disparity claims additionally require a significant subgroup/interaction term.

7. Figures and tables — main paper vs appendix

A 4-page short paper holds ~3 figures + ~2 tables.

Main paper

Item	Type	Shows	From
Fig 1	Partial-dependence, 2 panels	Δ WER vs speech-likeness and vs 2–8 Hz modulation , pooled over models	E1 / H1
Fig 2	Paired bars	Real vs scrambled on speech-like backgrounds @ 0 dB (the semantic gap)	E2 / H2–H3
Fig 3	Grouped bars	Disparate robustness: Δ WER by accent/gender, speech-like vs stationary, clean gap overlaid	E3 / H4
Table 1	Summary	model × {clean, noisy@+10, Δ , worst-background Δ , injection rate} for ASR & KWS	headline
Table 2	Regression	standardized descriptor coefficients per task	E1

Appendix

Item	Type	Shows
Fig A1	Lines	SNR dose-response per background cluster (monotonicity)
Fig A2	Stacked bars	Error/injection taxonomy (sub/del/ins/injection), clean vs noisy, per model
Fig A3	Bars	Steerability RER per model (if E4 run)
Fig A4	Scatter	Battery map: backgrounds in descriptor space, colored by harmfulness
Fig A5	Bars	Architecture view: Δ vs encoder-coupling tier (descriptive)
Table A1	Full	Per-background leaderboard (Δ per task)
Table A2	Checks	Manipulation checks: background-presence, Whisper WER-constancy, scramble-validity, determinism floor
Table A3	Fairness	Full subgroup × background-type Δ with CIs
Table A4	Prior art	Differentiation table (Appendix A)

If injection is the most striking result, promote Fig A2 into the main paper and demote the architecture view.

8. Compute budget (16 GB-aware arithmetic)

Block	Calc	Passes
E1 ASR	$100 \times 20 \text{ bg} \times 3 \text{ SNR} \times 5 \text{ models}$	30,000
E1 ASR clean + text-oracle	$100 \times 5 \times 2$	1,000
E1 KWS	$120 \times 20 \times 3 \times 5$	36,000
E1 KWS clean	120×5	600
E2 scrambled (ASR+KWS)	$8 \text{ bg} \times 220 \text{ items} \times 5 \text{ models @ } 0 \text{ dB}$	8,800
E3 controlled (SAA)	$90 \times 12 \text{ bg} \times 5 \text{ models @ } 0 \text{ dB}$	5,400
Core subtotal		$\approx 82 \text{ k}$
E4 steerability (stretch)	$220 \times 8 \text{ bg} \times 4 \text{ models @ } 0 \text{ dB}$	+7,040
Stretch: 2 extended models, reduced ASR+KWS	$220 \times 12 \text{ bg} \times 2$	+5,280
With stretches		$\approx 94 \text{ k}$

Wall-clock on 16 GB. Smaller batches than 24 GB → assume ~1.5-2 s/clip un-batched for 7B; ~82 k × 1.75 s ≈ 40 GPU-hours core ≈ **~2.5 inference days** at ~16 h/day; KWS batching shortens this. Comfortable inside 10 days.

Fallback knobs if a day slips: 4 models · SNR {+10, 0} · 16 backgrounds · skip E4 · E3 ecological-cut only.

9. Timeline (10 days)

Day	Work	Output
1	Env; download ESC-50, DEMAND (5 envs), MUSAN subset, NOISEX babble, LibriSpeech + Speech Commands subsets, SAA; stream Common Voice; build foreground item banks	sources + JSONL banks
2	Curate ~20 backgrounds; build 8 scrambled twins; extract descriptors; freeze prereg ; write the on-the-fly mixer + materialize() + diagnostics logger	battery + descriptors + prereg/ + mixer
3	5 model adapters (16 GB, 4-bit, understanding-only); manipulation checks on the inspection sample; determinism check	adapters + checks/
4–5	Run E1 (ASR + KWS) , 5 models (the bulk)	inference/ JSONL
6	Go/no-go gate ; run E2 scrambled + E3 SAA + text-oracle (+ E4 if ahead)	controls + fairness
7	Stretch (E4 / extended models) or start scoring	—
8	Scoring (WER/FAR/BIR/TIR); descriptor regression ; C-FAIR cuts	results/
9	Figures 1–3 (+appendix); disparity stats; CIs	plots
10	Write the 4-page draft; buffer	submission draft

Go/no-go (Day 6): if E1 shows neither a descriptor law beyond SNR nor a disparity signal, pivot to the **robustness-null** framing rather than padding tasks.

10. Risks

Risk	Mitigation
7B model OOMs on 16 GB	4-bit + understanding-only; per-model VRAM dry-run; 3B/edge swaps in the roster; free GPU between models.
Compute/time overrun	On-the-fly mixing (no I/O bottleneck) + KWS batching; fallback knobs; Day-6 gate.
Storage	Source corpora only (< 3 GB); stream Common Voice; never persist the stimulus set.
Reproducibility of backgrounds	All backgrounds are named public corpora; release exact source IDs + seeds; any clip regenerable via materialize() .
"On-the-fly = opaque"	Persisted inspection sample + deterministic regeneration + per-stimulus diagnostics CSV (§3.1).
Not novel vs RSA-Bench	Lead with descriptor law + scrambled control + fairness; cite/differentiate (Appendix A).
"Babble is just loud"	Scrambled twin + descriptor regression (SNR is a separate regressor) + text-oracle.
Descriptor collinearity	Report VIF; dissociate with NOISEX-babble vs MUSAN-stationary; prefer partial-dependence to raw coefficients.
Model quirks (Kimi ASR-default; speech-out)	Robust adapters; defensive parsing; Whisper-transcribe speech-out; log parse failures.

11. Expected outcomes

- **If H1-H4 hold:** a released **descriptor-annotated battery built from real public corpora**, a lightweight harness, a **generalizable law** of which sound properties break ASR/KWS, a **disparate-robustness** finding with an explanatory account, and a **scramble-proven semantic-injection** result — a concrete deployment warning for voice assistants in noisy, multi-speaker settings. On-theme for IMPACT-SPEECH.
- **If H0 holds:** a rigorously-controlled demonstration that current LALMs treat background as undifferentiated energy and degrade equitably — a trustworthy guarantee plus a reusable benchmark.

Minimum viable result: E1 (ASR+KWS) + the C-PROFILE regression + the E2 scrambled control + the E3 ecological cut.

Appendix A — Prior-art differentiation

Prior work	Does	Leaves open (our wedge)
RSA-Bench (2601.10384)	4 hand-built scenarios, source-count K=1–4, fixed energy (no SNR sweep) , 6 tasks; vocal-like > mechanical; denoising paradox	scenarios re-import event→scene mapping; no descriptor law, no SNR dose-response, no fairness, no scrambled control
Do LLM Decoders Listen Fairly? (2604.21276)	LLM-decoder ASR bias across accent/gender, incl. under degradation	ASR only; generic noise; no characterization of which backgrounds widen the gap ; no KWS/injection
FairLENS (2405.13166)	disparate ASR degradation under noise (law-enforcement)	classical ASR, not LALMs; coarse noise; single task
VoiceBench (2410.17196)	voice-assistant benchmark incl. a noise track	noise is one knob; no fairness/descriptor/injection
WildSpeech-Bench (2506.21875)	natural-conversation QA + background-speech/ESC-50 augmentation	robustness sub-track; no descriptor law/fairness/injection
AudioTrust (2505.16211)	trustworthiness incl. background-conversation/env-sound scenarios	coarse buckets; no per-descriptor law, no disparity, no SNR sweep
VocalBench-DF (2510.15406)	speech-LLM robustness to disfluency/overlap	speaker-side, not environmental-background characterization or fairness
Object-hallucination / distractor in LALMs (Kuan et al. 2024; 2025)	LALMs hallucinate absent sounds / distracted in audio-QA	audio-QA reasoning, not a cross-task background-robustness <i>failure taxonomy</i> tied to descriptors + demographics

Verdict: "noise hurts LALMs" is occupied. The defensible contribution is the **descriptor law + disparate robustness + scramble-proven semantic injection**, with controls (SNR sweep, scrambled twin, text-oracle) the closest competitor omits.

Appendix B — Glossary

- **Interference battery** — the shared, manipulated variable: ~20 real background recordings (ESC-50, DEMAND, MUSAN, NOISEX) each with a descriptor vector.
- **Descriptor** — a measurable property of a background (speech-likeness, linguistic content, 2–8 Hz modulation, spectral overlap, stationarity, onset density, loudness).
- **Scrambled twin** — phase-scrambled / spectrally-matched version of a background (same spectrum & energy, no structure or meaning).
- **BIR / TIR** — Background-Injection Rate (ASR) / Trigger-Injection Rate (KWS): how much background content leaks into outputs.
- **Robustness Gap / DRI** — disparity in the *noise penalty* across demographic subgroups.
- **RER** — Residual-Effect Ratio: how much of the effect survives an "ignore background" instruction.
- **materialize(row)** — regenerate the exact waveform for any manifest row, on demand, for inspection.
- **On-the-fly mixing** — generating noisy waveforms in RAM at inference from seeds, never persisting them, while staying fully inspectable via the three mechanisms in §3.1.